

$$a + b = 10$$

$$2a + 3b = 15$$

$$3.4a + 48.96b = 122$$

Only Numbers allowed

or can say max power = 1
degree = 1

or can say
no variable multiplied

Straight line
↓
Linear Regression

$$a^2 + b^2 = 10$$

$$31(a) + b^2 = 15$$

$$ab^2 + \frac{b}{a} - \frac{3}{b} = \log b$$

powers > 1,

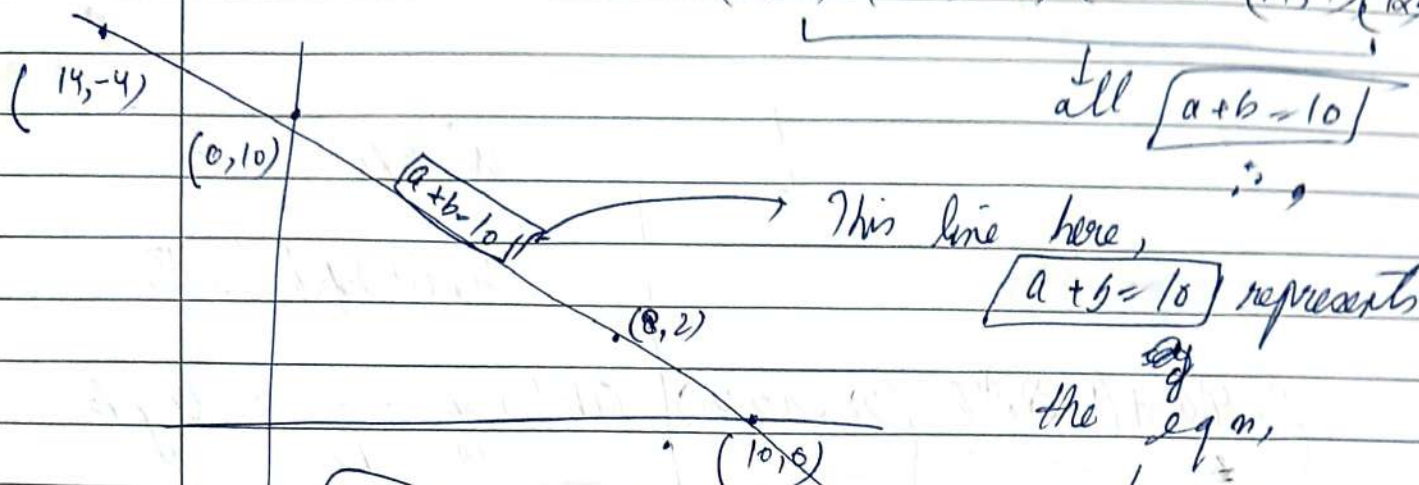
or
var. multiplied with each other

↓
Curved line (not straight)

↓
Neural network, deep learning

eg

$$a + b = 10 \rightarrow \text{sol}^n \rightarrow (10, 0), (0, 10), (8, 2), (14, -4), (-2, 2)$$



every point on the line is the solⁿ to the eqⁿ

so by slope $m = \frac{dy}{dx} = \frac{1}{1} = 1$

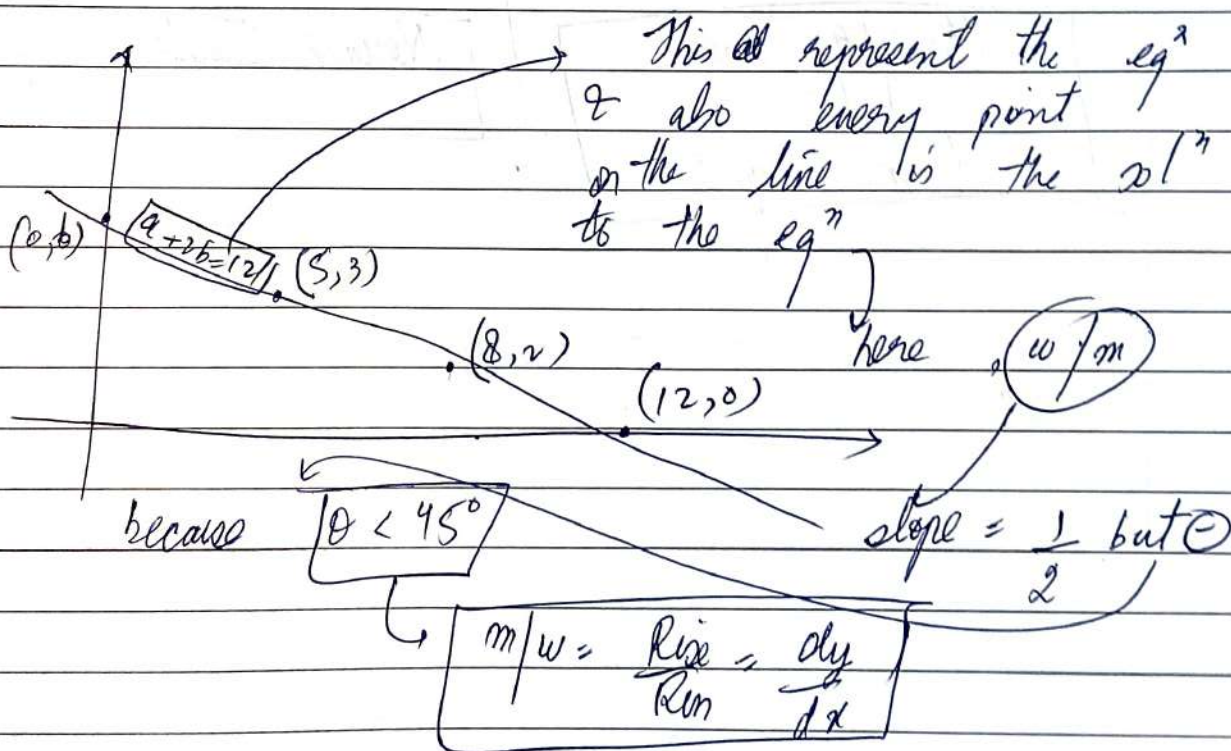
or (w)

Now,

$$a + 2b = 12$$

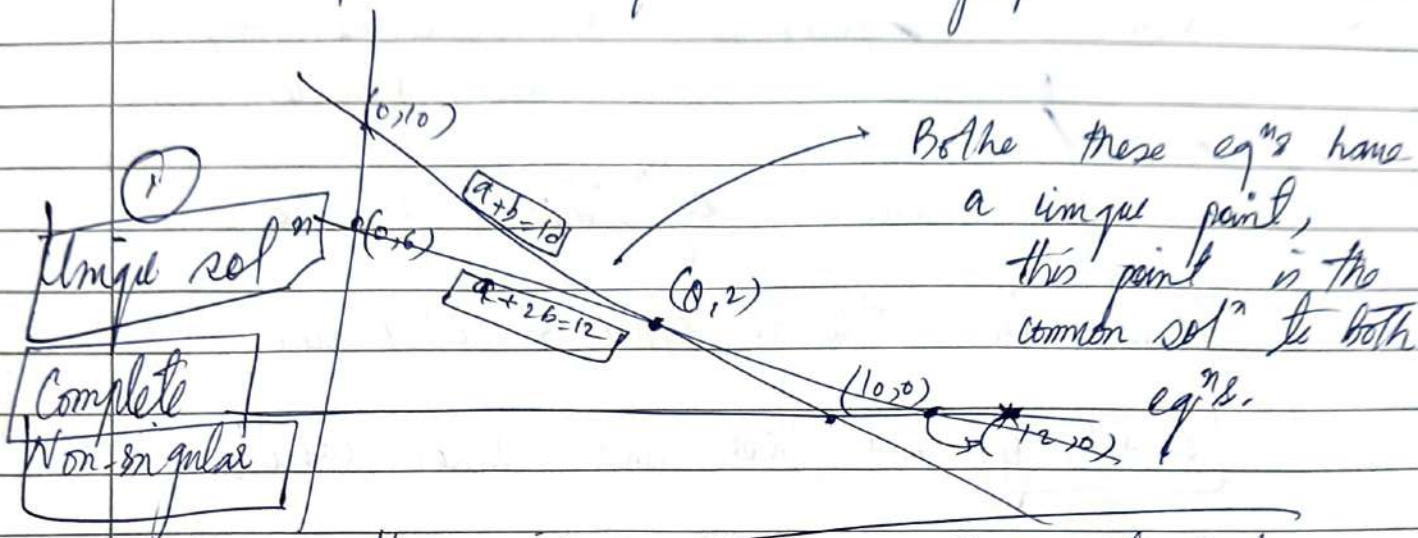
$$\frac{dy}{dx} = \frac{1}{2} = 0.5$$

$$\text{sol}^n = (0, 6), (8, 2), (5, 3), (12, 0)$$



System solns

Now, plot these eqⁿ is same graph



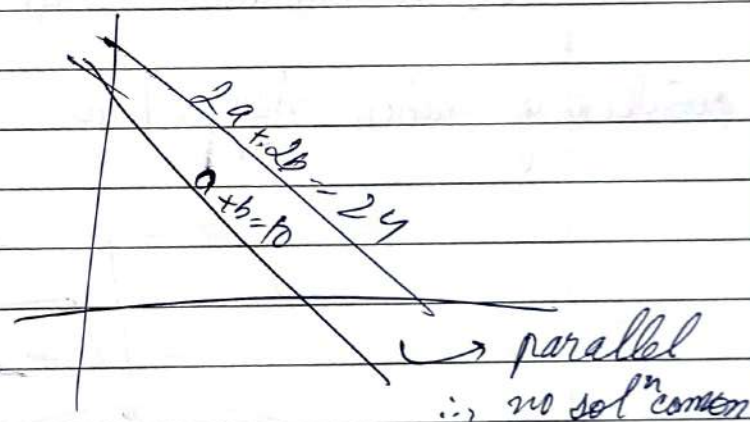
trying that with $a+b=10$ & $2a+2b=20$

② Infinitely many solⁿs
Therefore every same imp lines & overlap each other
as discussed before
Redundant
~~Non-singular~~

(eg) $a+b=10, 2a+2b=20$

③ No solⁿ

Contradictory
~~Non-singular~~



④ finding if a system is sing or non-sing

If the eqns have diff slopes $\rightarrow$ they must intersect & therefore have a unique solⁿ

If the slopes are equal

Coincides
 $\rightarrow$ solⁿ

Parallel
No solⁿ

_ / _ / _

→ Non system of 3 linear eq^{ns} in 3 variables

so for each eqⁿ, there is a plane in the 3D space, Just like

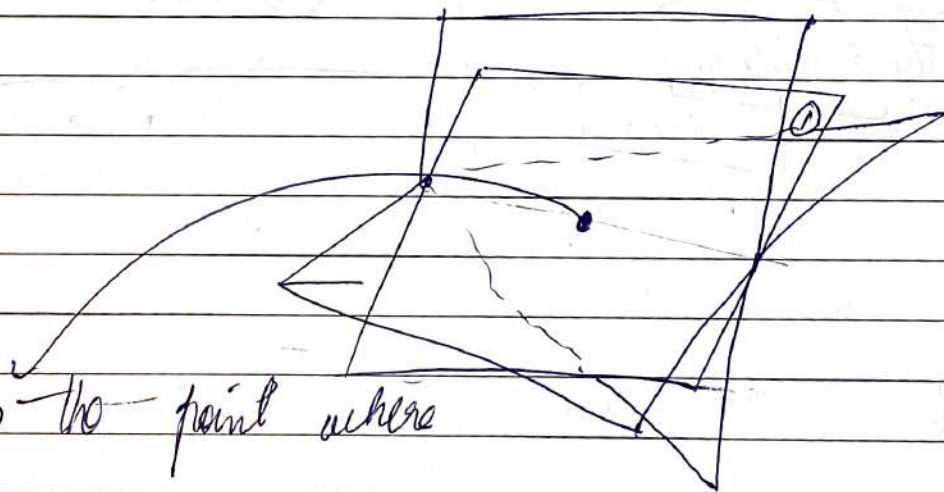
$a + b = c$ → these had lines which

meet or not meet in diff ways,

3 variables have also that type, ↓

① All 3 planes from the 3 eq^{ns} intersect at only one point simultaneously

possible only when they have diff slopes & no are parallel.



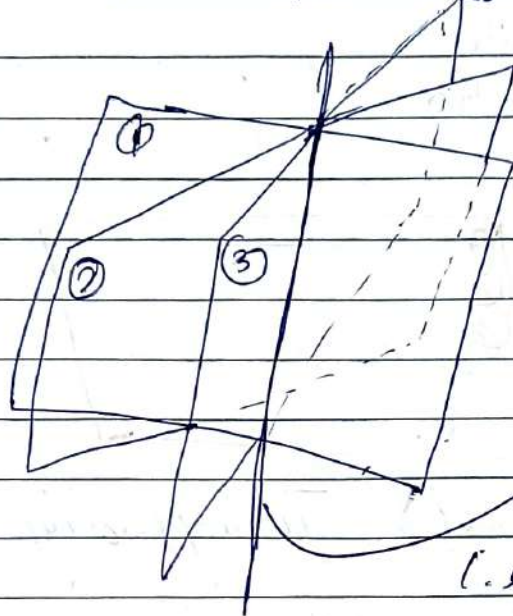
here is the point where

all these planes meet simultaneously, so

that is the only unique solⁿ

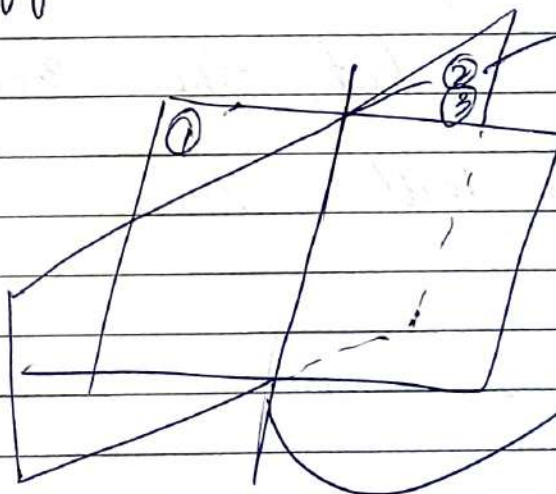
of the system of eq^{ns}.

② here the three planes must intersect at one line



here, all 3 planes meet on a line
i.e., ∞ solⁿs.

Note - It is not important for all three planes to be different as one case here can have two same planes & 1 diff

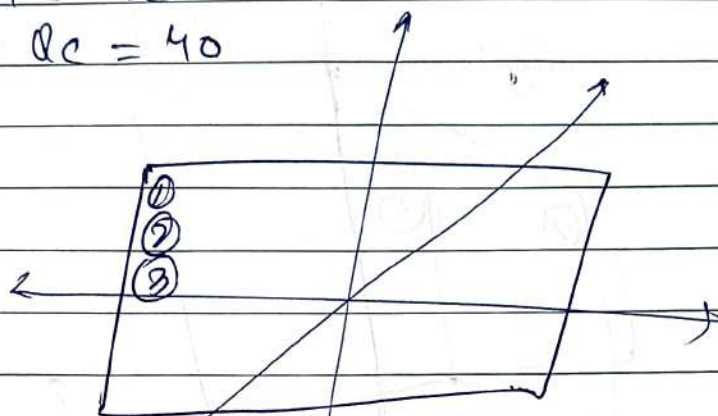


2 planes coincide

same ∞ solⁿs to the system of eqⁿ

③ system having all planes identical

e.g) $a + 2b + 2c = 10$
 $2a + 4b + 4c = 20$
 $4a + 8b + 8c = 40$



∞ no. of solⁿs corresponding to every point of the plane.

→ ④ no solⁿ if all three doesn't intersect at a point / line.

Even if 2 planes are identical & intersect but the 3rd doesn't, the system will have no solⁿs.

• Determinant

↳ Tells if a matrix is singular

$$\boxed{D=0}$$

Singular

$$D \neq 0$$

Non Singular

Linear Dependence b/w rows

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \rightarrow 2 \times 2$$

Linear indep

$$k = ?$$

Not Non Sing

$$D = 2 - 1 = 1$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \rightarrow 2 \times 2$$

Linear dep.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$D = 2 - 2 = 0$$

$$(a, b) \times k = (c, d) \text{ Sing}$$

for $a \begin{bmatrix} 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$D = 1 \times 4 - 3 \times 2$$

$$\boxed{D = 4 - 6 = -2}$$

for (3×3) matrix

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$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

$$D =$$

$$D = a(ei - fh) - b(di - fg) + c(dh - eg)$$

$$D = a \times \det \begin{bmatrix} e & f \\ h & i \end{bmatrix} - b \times \det \begin{bmatrix} d & f \\ g & i \end{bmatrix} + c \times \det \begin{bmatrix} d & e \\ g & h \end{bmatrix}$$

Q) Det

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 3 & 3 & 3 \end{bmatrix} \det = 1(3-0) - 0(\dots) + 1(0-3)$$

$\therefore -3 = 0$ Singular matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & -1 \end{bmatrix} \Rightarrow 1(-1-0) - 1(-1-0) + 1(0)$$

$$-1 + 1 = 0 \text{ } \Rightarrow \text{sing m.}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \det = 1(6-0) - 1(0-0) + 1(0)$$

$$\det = 6 \text{ Non singular matrix}$$

This is an upper Δ matrix

as all other elements are zero

& a short trick is that:

For upper triangular matrix $D = \text{product of main diagonal}$

$1 \times 2 \times 2 = 6 \rightarrow$ They can be singular or non-singular as well.

as in

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \det = 0$$

Non-singular